Do You Remember? Overcoming Catastrophic Forgetting for Fake Audio Detection

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Introduction

Motivation

Current fake audio detection algorithms have achieved promising performances on most datasets. However, their performance may be significantly degraded when dealing with audio of a different dataset. The orthogonal weight modification to overcome catastrophic forgetting does not consider the similarity of genuine audio across different datasets.

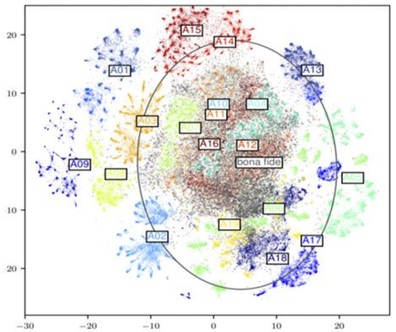

Figure 1. The feature distribution of bona-fide and spoof audio.[1]

Contribution

- We propose a regularized adaptive weight modification (RAWM) algorithm to overcome catastrophic forgetting. Our method can modify weight's direction according to the number of genuine and fake audio.
- Although our method is inspired by the feature distribution similarity in fake audio detection, it can also be used in other related tasks.

[1] Ma, Haoxin, et al. "Continual learning for fake audio detection." arXiv preprint arXiv:2104.07286 (2021).

Proposed Method

On most fake audio detection datasets, under the same acoustic conditions, feature distributions of genuine audio are relatively more concentrated than the fake.

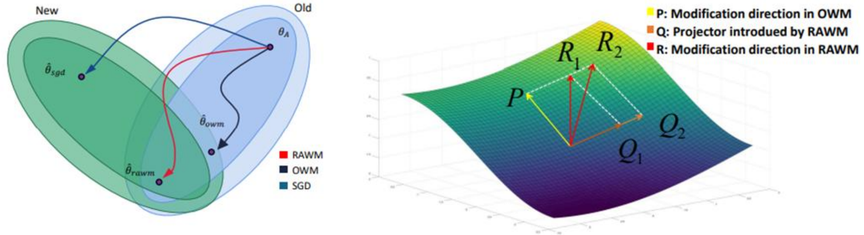

Figure 2. Schematic of SGD, OWM, and RAWM

Adaptive Weight Modification

We introduce a square matrix Q as a projector that is orthogonal to the P proposed by the OWM algorithm. The modified direction R of weights, introduced in our paper, can be written as:

$$R = P_{norm} + mQ_{norm}$$

where m is a constant; P_{norm} and Q_{norm} are identity matrices normalized by P and Q, respectively.

Regularization

There are also a few datasets where genuine audio is collected from quite different acoustic conditions compared with others. As for these utterances, we use the teacher-student framework as a regularization term.

Regularized Adaptive Weight Modification

Considering a continual learning situation, the BP process of regularized adaptive weight modification can be written as:

$$W_l(i,j) = W_l(i-1,j) + \gamma(i,j)\Delta W_l^{BP}(i,j) \qquad j=1$$

$$W_l(i,j) = W_l(i-1,j) + (1-\eta)R_l(j-1)\gamma(i,j)\Delta W_l^{BP}(i,j) + \eta\Delta W_{l_{reg}}^{BP}(i,j) \qquad j>1$$

Experimental Results

(a)

Method	S	T_1
Baseline	0.258	24.532
Train-on-All	0.406	0.201
Fine-tune	7.324	0.510
EWC	2.832	0.570
OWM	2.448	0.540
LwF	3.123	0.343
DFWF	1.849	0.689
RAWM(Ours)	**0.666**	**0.247**

(b)

Method	S	T_2
Baseline	0.258	46.503
Train-on-All	0.965	2.498
Fine-tune	8.755	5.647
EWC	3.494	6.289
OWM	3.086	6.432
LwF	4.234	4.566
DFWF	1.874	7.355
RAWM(Ours)	**1.237**	**3.721**

(c)

Method	S	T_3
Baseline	0.258	91.473
Train-on-All	2.740	2.160
Fine-tune	20.976	4.978
EWC	8.039	5.615
OWM	8.130	5.065
LwF	6.453	4.998
DFWF	4.324	6.275
RAWM(Ours)	**4.942**	**3.249**

(d)

Method	S	T_1	T_2	T_3
Baseline	0.258	24.532	46.503	91.473
Train-on-All	1.324	0.561	3.579	2.008
Fine-tune	7.068	2.841	5.674	4.543
EWC	5.569	2.444	6.510	5.129
OWM	4.083	2.167	6.480	5.472
LwF	2.714	1.621	4.875	4.325
DFWF	3.476	3.735	7.345	6.114
RAWM(Ours)	**1.508**	**0.641**	**3.850**	**3.163**

Table 1. The EER(%) of our method compared with various methods for fake audio detection.

Continual Learning Methods	Acc on the evaluation set of each experience									
	Exp_1	Exp_2	Exp_3	Exp_4	Exp_5	Exp_6	Exp_7	Exp_8	Exp_9	Exp_{10}
Replay	**94.34**	**93.64**	**94.34**	**95.15**	**94.75**	**94.55**	**94.34**	**94.34**	**95.35**	**96.06**
Fine-tune	87.68	90.00	91.11	91.82	90.40	89.90	90.30	90.61	90.61	93.33
EWC	84.04	84.95	85.86	87.07	85.66	85.56	86.97	86.16	85.76	87.78
LwF	88.59	88.89	87.27	90.51	87.68	87.78	87.47	87.47	88.79	88.48
GDF	91.11	91.62	88.38	91.01	88.79	89.19	90.20	87.68	90.10	90.30
CWR	90.71	91.72	90.71	91.52	89.49	90.91	91.62	90.71	91.82	93.74
OWM	91.62	92.12	**91.82**	93.64	91.72	**92.42**	92.22	**92.32**	92.42	95.05
RAWM (Ours)	**92.12**	**92.53**	91.41	**93.74**	**91.82**	**92.42**	**92.53**	92.22	**92.53**	**95.25**

Table 2. The Acc(%) of our method compared with various methods for image recognition.

We compared our method with several wide-used continual learning methods for fake audio detection. Among the results of our experiments, our method achieves the best learning and overcoming forgetting.

Conclusion

- As illustrated in experimental results, our method shows promising performance on fake audio detection, emotion recognition, and image recognition, even training on a few samples.
- We have yet to study how to make the model learn the weight direction gradually in the process of training on new datasets without any constraint, and exploring generalization to related tasks will form the focus of our future studies.